RESUME SCREENING USING NLP

A PROJECT REPORT

Submitted by

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Abstract

The rapid increase in job applications has made manual resume screening inefficient and time-consuming for organizations. To overcome this, the proposed system introduces an automated Resume Screening System using Natural Language Processing (NLP) to efficiently filter and rank candidates based on job requirements. The system processes unstructured resume and job description data by converting it into structured text and applying preprocessing techniques such as noise removal, stopword elimination, tokenization, and normalization. It then uses TF-IDF for feature extraction to transform text into numerical vectors, enabling effective comparison. Cosine similarity is applied to measure the relevance between resumes and job descriptions, while machine learning models like Multinomial Naive Bayes and K-Nearest Neighbors are used for classifying resumes into specific job roles. The system outputs a ranked list of candidates with similarity scores, helping recruiters quickly identify suitable applicants. Developed using Python with libraries such as Scikit-learn, NLTK, and Streamlit, the system provides an interactive interface and significantly reduces manual effort, improves efficiency, and minimizes bias, offering a scalable solution for modern recruitment processes

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CHAPTER 1

INTRODUCTION

1.1 Introduction

In today's recruitment process, companies receive thousands of resumes for a single job opening. Manually screening these resumes is time-consuming, inefficient, and prone to human bias. To solve this problem, this project proposes an automated Resume Screening System using Natural Language Processing (NLP). The system extracts relevant information such as skills from resumes and ranks candidates based on how well they match a given job description. By using techniques like TF-IDF (Term Frequency–Inverse Document Frequency) and text similarity, the system ensures faster, more accurate, and unbiased candidate selection.

1.2 Problem Statement:

Manual resume screening presents several significant challenges in modern recruitment processes. It is highly time-consuming, as recruiters must go through a large number of resumes manually. This process is also prone to human errors and bias, which can affect the fairness and accuracy of candidate selection. Additionally, handling a large volume of resumes becomes difficult, especially for organizations receiving hundreds or thousands of applications. Finally, it is challenging to precisely match candidates with job requirements, as manually identifying relevant skills and qualifications from unstructured resume data is complex and inefficient.

1.3 Objectives

The primary objective of this project is to develop an automated resume screening system using Natural Language Processing (NLP) techniques to improve the efficiency and accuracy of the recruitment process. The system aims to automatically process and analyze large volumes of resumes by extracting relevant information such as skills and key features from unstructured text data. It focuses on converting textual content into meaningful numerical representations and computing the similarity between resumes and job descriptions to identify the most suitable candidates. Additionally, the project seeks to rank candidates based on their relevance scores, enabling recruiters to quickly shortlist the best matches. Overall, the objective is to reduce manual effort, minimize bias, and provide a faster, scalable, and reliable solution for candidate selection.

CHAPTER 2

SYSTEM SPECIFICATION

2.1 Hardware Requirements

The hardware requirements for the proposed Resume Screening System are minimal, as the system is designed to run efficiently on standard computing devices. The following configuration is sufficient for smooth execution:

- **Processor:** Intel Core i3 or higher
- **RAM:** Minimum 4 GB (8 GB recommended for better performance)
- **Storage:** At least 500 MB free disk space
- **System Type:** 64-bit operating system
- **Input Devices:** Keyboard and Mouse
- **Output Devices:** Monitor/Display Screen

2.2 Software Requirements

The system is developed using Python and requires the following software components for execution:

- **Operating System:** Windows 10/11, Linux, or macOS
- **Programming Language:** Python (version 3.8 or above)
- **Development Environment:** Visual Studio Code / Jupyter Notebook
- **Web Framework:** Streamlit (for building the user interface)

Required Python Libraries:

- **Pandas:** For data manipulation and handling datasets
- **NumPy:** For numerical computations
- **Scikit-learn:** For machine learning algorithms and TF-IDF vectorization
- **NLTK:** For natural language processing tasks such as tokenization and stopword removal
- **Matplotlib / Plotly:** For data visualization
- **WordCloud:** For generating word cloud visualizations
- **pdfplumber (optional):** For extracting text from PDF resumes

CHAPTER 3

SOFTWARE DESCRIPTION

3.1 Frontend

The frontend of the system is developed using **Streamlit**, a Python-based web application framework. It enables the rapid development of interactive web interfaces without requiring separate frontend technologies such as HTML, CSS, or JavaScript.

Streamlit provides components such as text input fields, file uploaders, buttons, and dynamic output displays like tables and charts. It allows users to easily interact with the system by uploading resumes, entering job descriptions, and viewing results such as classification outputs and ranked candidates.

3.2 Backend

The backend of the system is implemented using **Python (version 3.8 or above)**. Python is chosen due to its simplicity and extensive ecosystem of libraries for data science and machine learning.

The backend handles all core functionalities, including data preprocessing, feature extraction, model training, prediction, and ranking. It integrates multiple libraries to efficiently process textual data and generate meaningful outputs.

3.3 Libraries and Technologies Used

The system utilizes several important libraries and tools, which are described below:

- **Pandas & NumPy:**
  These are fundamental data manipulation libraries. Pandas is used to load and manage datasets such as *ResumeDataSet.csv*, perform data preprocessing, and handle structured data. NumPy supports numerical operations and efficient array handling.
- **Scikit-learn:**
  This is the primary machine learning library used in the system. It provides essential components such as:
  - **TfidfVectorizer:** Converts textual data into numerical feature vectors.
  - **Multinomial Naive Bayes (MNB):** Used for classification of resumes into categories.
  - **K-Nearest Neighbors (KNN):** Used for classification based on similarity.
  - **OneVsRestClassifier:** Enables handling of multi-class classification problems.
  - **cosine_similarity:** Calculates similarity between resume and job description for ranking.
- **NLTK (Natural Language Toolkit):**
  This library is used for text preprocessing tasks such as tokenization, stopword

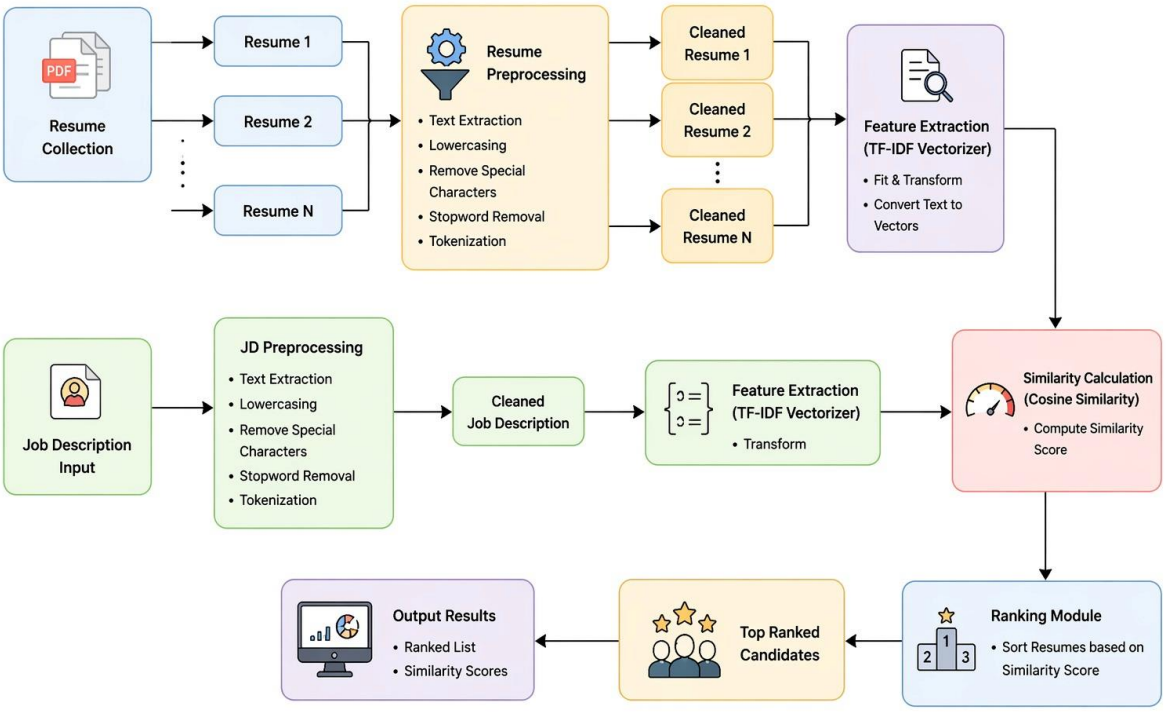
removal, and lemmatization. It helps in cleaning and normalizing textual data before further processing.

- **Plotly & Matplotlib / WordCloud:**
  These libraries are used for data visualization. Plotly generates interactive charts such as bar graphs and pie charts, while WordCloud creates visual representations of frequently occurring words in the dataset.
- **pdfplumber (Optional):**
  This library is used to extract text from PDF files, allowing the system to process resumes uploaded in PDF format.

CHAPTER 4

PROJECT DESCRIPTION

4.1 System Architecture



Resume Preprocessing

The Resume Preprocessing module processes the raw resume data to make it suitable for analysis. In this stage, text is extracted and converted into a standardized format by applying operations such as lowercasing, removal of special characters, elimination of stopwords, and tokenization. This ensures that the data is clean, consistent, and ready for feature extraction.

Job Description Input

The Job Description Input module accepts the job requirements provided by the recruiter. This serves as a reference document against which all resumes will be evaluated. It acts as the baseline for matching candidate profiles.

Job Description Preprocessing

The Job Description Preprocessing module performs similar cleaning operations as applied to resumes. The job description text is processed through steps such as text extraction, lowercasing, removal of unwanted characters, stopword removal, and tokenization. This ensures uniformity between resume data and job description data.

Feature Extraction (TF-IDF Vectorizer)

The Feature Extraction module converts the cleaned textual data into numerical form using the TF-IDF technique. It assigns importance weights to words based on their frequency and

relevance within the document. This transformation allows both resumes and job descriptions to be represented as vectors in a common feature space.

Similarity Calculation (Cosine Similarity)

The Similarity Calculation module compares the numerical vectors of resumes with the job description vector. Cosine similarity is used to compute a similarity score that indicates how closely a resume matches the job requirements. The score typically ranges from 0 to 1, where higher values indicate better matches.

Ranking Module

The Ranking module sorts all resumes based on their similarity scores in descending order. This step helps in identifying the most relevant candidates by prioritizing those with higher similarity values. It simplifies the selection process for recruiters.

Top Ranked Candidates

The Top Ranked Candidates module filters and highlights the most suitable candidates from the ranked list. It identifies candidates with the highest scores, making it easier to shortlist applicants for further evaluation.

Output Results

The Output Results module displays the final results of the system. It presents a ranked list of candidates along with their similarity scores, enabling recruiters to quickly analyze and select the best candidates for the job.

4.2 Workflow of the Project

1.Application Startup: The user launches the app. The system executes

app.py, sets up the page configuration, injects CSS, and attempts to download necessary NLTK data silently.

2. Data Ingestion & Exploration: The user navigates to the Data Analysis page. The system reads ResumeDataSet.csv into a Pandas dataframe,

applies the cleaning function to all resumes, encodes the text categories

into integers using LabelEncoder, and caches this dataframe. Charts are

generated from this cached data.

3. Model Training: The user navigates to the Train Models page and initiates training.

• The system takes the cached dataframe and uses TfidfVectorizer

to create the feature matrix X.

• Data is split into 75% training and 25% testing sets.

• Both MultinomialNB and KNeighborsClassifier (wrapped in

OneVsRestClassifier) are trained on the training set.

• The system evaluates them against the test set, generates accuracy scores, and saves the models and vectorizer to disk.

4. Prediction Phase: The user goes to the Predict page and inputs text.

• The system reads the saved .pkl files from disk into memory.

• The raw input is passed through the text cleaner.

• The cleaned string is transformed by the loaded TfidfVectorizer into a vector.

• The selected model's .predict() method is called on the vector to get the category ID.

• The ID is mapped to a human-readable string via the CATEGORY_MAPPING dictionary and displayed to the user.

5. Ranking Phase: The user uploads resumes and a JD.

• Text is extracted from uploaded files (using pdfplumber if applicable).

• All texts are cleaned.

• A new TF-IDF matrix is created containing the JD at index 0 and resumes at subsequent indices.

• cosine_similarity is calculated between index 0 and all other indices.

• Results are sorted descending by score and displayed alongside a generated bar chart

4.3 Model Details

• Model Types:

– **Multinomial Naive Bayes (MNB):** A probabilistic learning method based on Bayes' theorem. It calculates the probability of each category given the TF-IDF weights of the words in the resume, assuming word occurrences are independent.

– K-Nearest Neighbors (KNN) with OneVsRest: A nonparametric method that plots the resume in a multi-dimensional space and looks at the 'K' closest resumes from the training data. OneVsRest breaks the 25-class problem into 25 separate binaryclassification problems (e.g., "Is Java Developer" vs "Is NOT Java Developer") to improve accuracy.

• **Training Process:** The models are trained on the preprocessed text from ResumeDataSet.csv. The text is vectorized using TF-IDF (ignoring English stopwords). The target labels are numerically encoded. The data is split 75/25 for training/testing.

• **Input Preprocessing:** Relies heavily on aggressive regex cleaning to remove noise (URLs, symbols) and lemmatization to reduce feature dimensionality (ensuring “develop”, “developer”, and “developing” map to the same root).

• **Limitations:**

– The models are “bag-of-words” based, meaning they entirely ignore word order and context. “Manager of developers” and “Developer of managers” look identical to the model.

– They are highly dependent on the vocabulary of the training dataset. Completely novel skills or job titles not seen during training will be ignored or misclassified.

4.4 Key Design Decisions

• **Streamlit for UI**: Chosen to eliminate the need for a separate frontend codebase (like React/Vue) and API layer (like FastAPI/Flask). It allows the entire ML pipeline and UI to be written in a single Python file, maximizing development speed.

• **TF-IDF vs. Word Embeddings**: TF-IDF was chosen over advanced embeddings (like BERT or Word2Vec) because it is computationally inexpensive, requires no GPU, and trains in seconds on a standard CPU while still providing strong baselines for document classification.

• **Pickling Models:** To ensure a responsive user experience, models are trained once and serialized to disk. This design decision prevents the system from locking up for 10-20 seconds every time a user refreshes the page or requests a prediction.

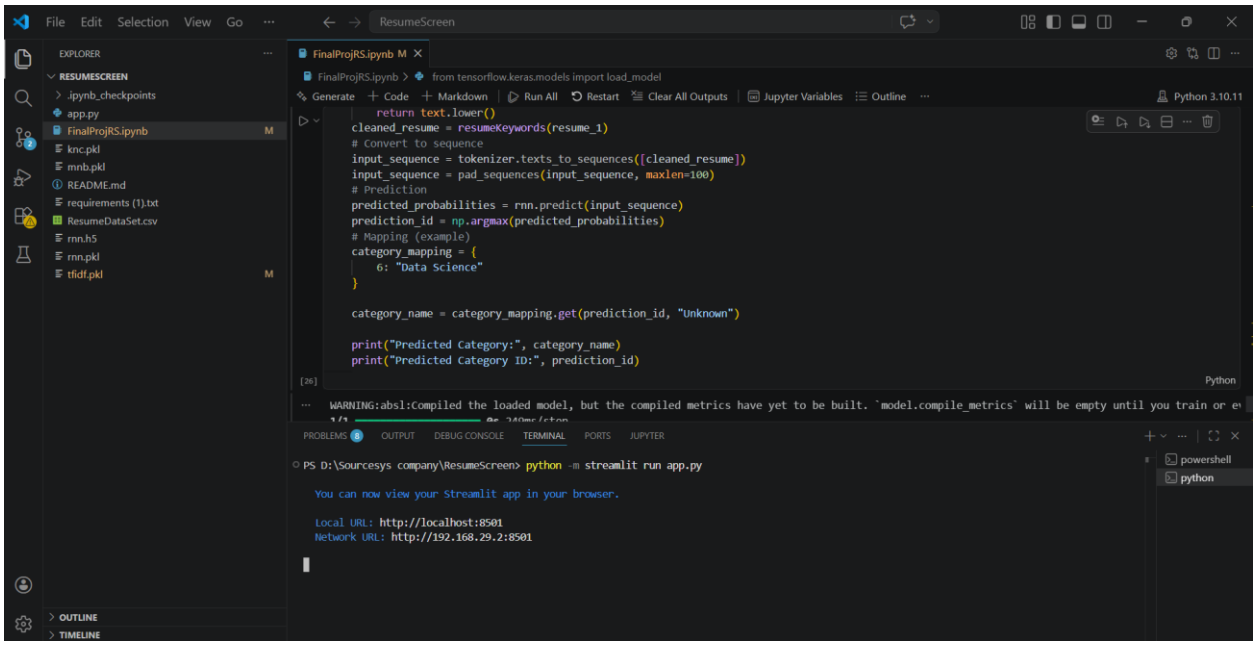
• **Dynamic Vectorization for Ranking:** During the ranking process, the system dynamically fits a new TF-IDF vectorizer specifically on the uploaded JD and resumes, rather than using the globally trained one. This ensures that unique keywords specific to that particular job description are properly weighted in the cosine similarity calculation, rather than being ignored if they didn’t exist in the original training data.

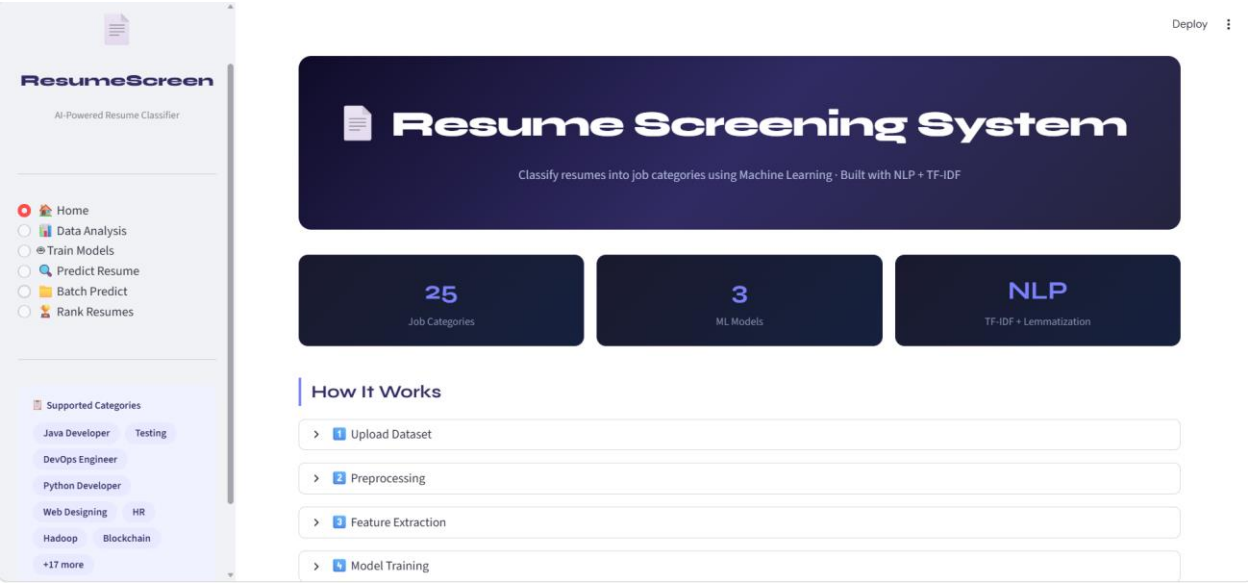
CHAPTER 5

OUTPUT AND RESULT ANALYSIS

5.1 Result Analysis:

OUTPUT:

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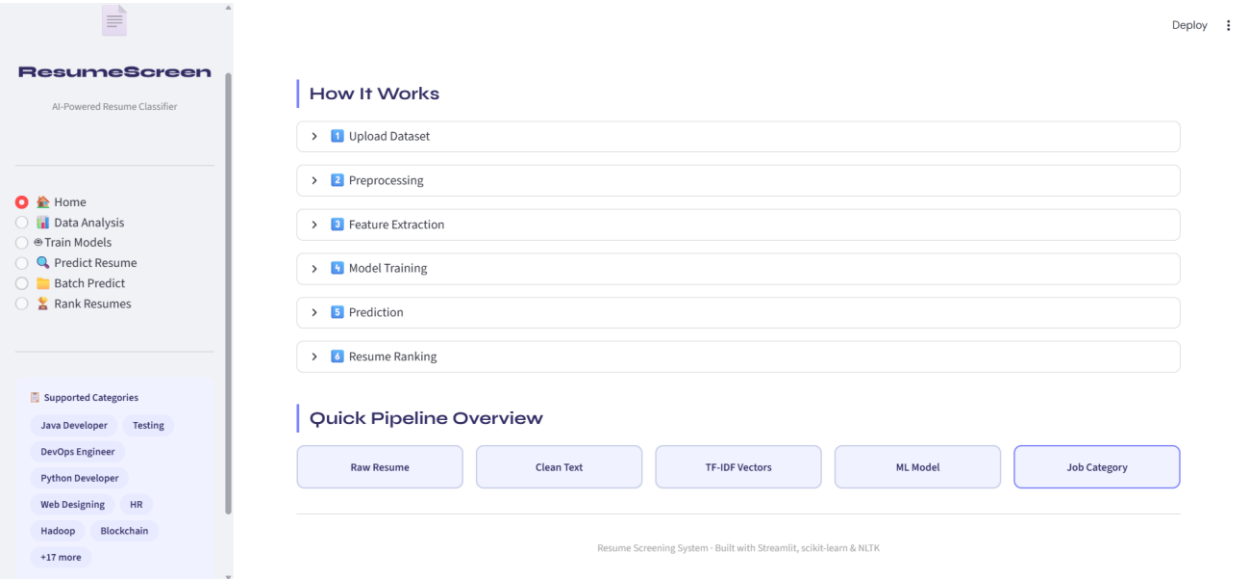
+17 more
Deploy
Resume Screening System
Classify resumes into job categories using Machine Learning · Built with NLP + TF-IDF
25
Job Categories
3
ML Models
NLP
TF-IDF + Lemmatization
How It Works
1 Upload Dataset
2 Preprocessing
3 Feature Extraction
4 Model Training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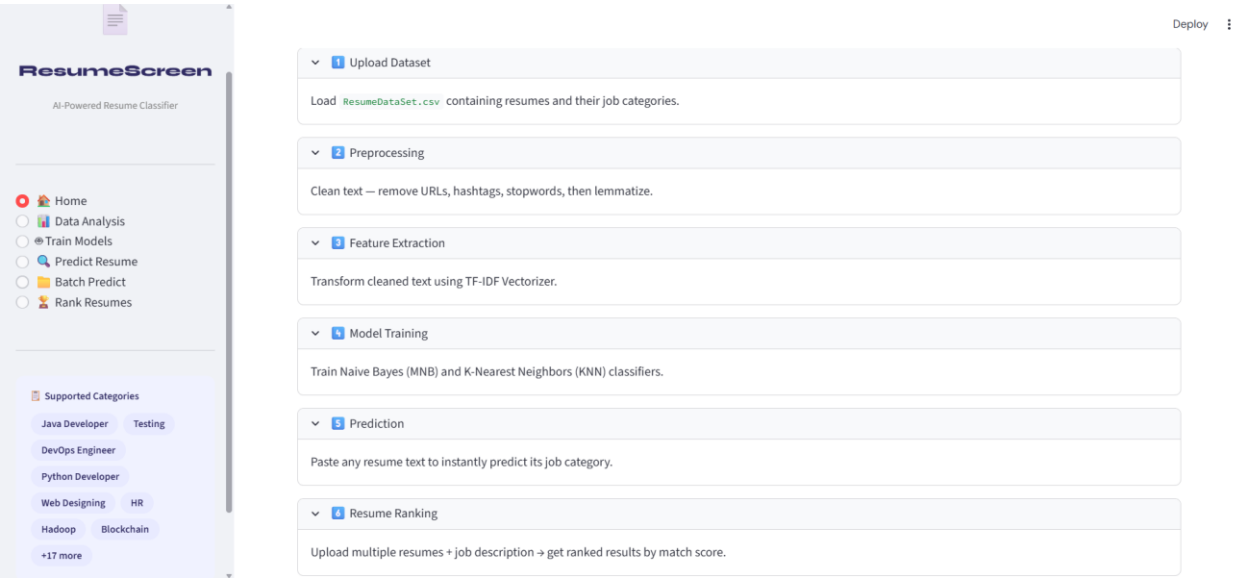
5 Prediction
6 Resume Ranking
Quick Pipeline Overview
Raw Resume
Clean Text
TF-IDF Vectors
ML Model
Job Category
Resume Screening System · Built with Streamlit, scikit-learn & NLTK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Deploy
1 Upload Dataset
Load ResumeDataSet.csv containing resumes and their job categories.
2 Preprocessing
Clean text — remove URLs, hashtags, stopwords, then lemmatize.
3 Feature Extraction
Transform cleaned text using TF-IDF Vectorizer.
4 Model Training
Train Naive Bayes (MNB) and K-Nearest Neighbors (KNN) classifiers.
5 Prediction
Paste any resume text to instantly predict its job category.
6 Resume Ranking
Upload multiple resumes + job description → get ranked results by match score.

ResumeScreen

AI-Powered Resume Classifier

- Home
- Data Analysis
- Train Models
- Predict Resume
- Batch Predict
- Rank Resumes

Supported Categories

Java Developer | Testing | DevOps Engineer | Python Developer | Web Designing | HR | Hadoop | Blockchain | +17 more

Deploy

Resume Screening System

Classify resumes into job categories using Machine Learning · Built with NLP + TF-IDF

Dataset Overview

Total Resumes	Unique Categories	Null Values
962	25	0

Raw Data | Bar Chart | Pie Chart | Word Cloud

	Category	Resume
0	Data Science	Skills * Programming Languages: Python (pandas, numpy, scipy, scikit-learn, matplotlib), Sql, Java, JavaScript/JQuery. * Mach
1	Data Science	Education Details May 2013 to May 2017 B.E UIT-RGPV Data Scientist Data Scientist - Matelabs Skill Details Python- Exprienc
2	Data Science	Areas of Interest Deep Learning, Control System Design, Programming in-Python, Electric Machinery, Web Development, Analy

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3	Data Science	Skills â¢ R â¢ Python â¢ SAP HANA â¢ Tableau â¢ SAP HANA SQL â¢ SAP HANA PAL â¢ MS SQL â¢ SAP Lumira â¢ (
4	Data Science	Education Details MCA YMCAUST, Faridabad, Haryana Data Science internship Skill Details Data Structure- Exprience - Les
5	Data Science	SKILLS C Basics, IOT, Python, MATLAB, Data Science, Machine Learning, HTML, Microsoft Word, Microsoft Excel, Microsoft Powe
6	Data Science	Skills â¢ Python â¢ Tableau â¢ Data Visualization â¢ R Studio â¢ Machine Learning â¢ Statistics IABAC Certified Data S
7	Data Science	Education Details B.Tech Rayat and Bahra Institute of Engineering and Biotechnology Data Science Data Science Skill Detail
8	Data Science	Personal Skills â¢ Ability to quickly grasp technical aspects and willingness to learn â¢ High energy levels & Result oriented
9	Data Science	Expertise â¢ Data and Quantitative Analysis â¢ Decision Analytics â¢ Predictive Modeling â¢ Data-Driven Personalizatic

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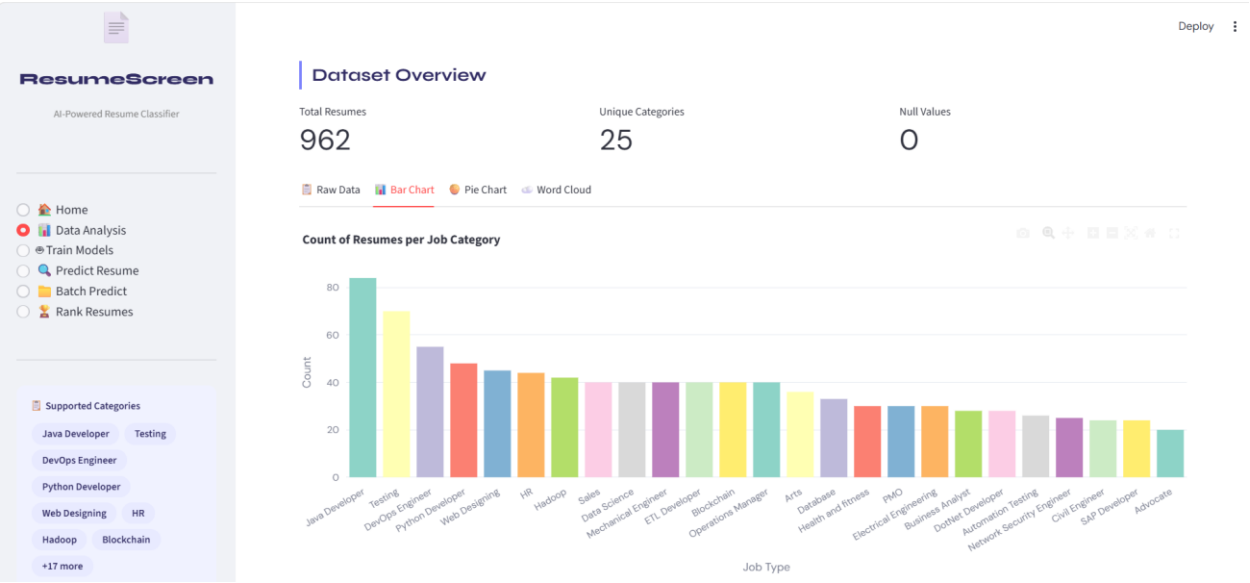
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Count
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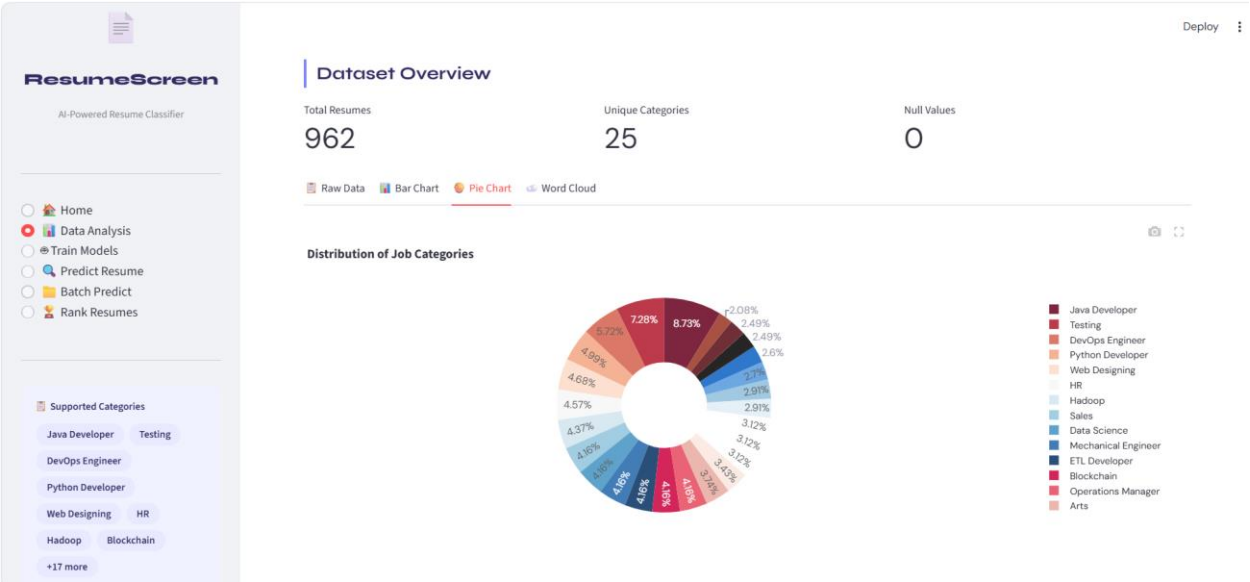
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Arts

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exprience monthscompany
role responsibility
company description
detail january
maharashtra state
college engineering
java developer
shell scripting
project description
year monthscompany
mumbai maharashtra
state board
management system
business requirement
project management
project name
machine learning
technology used
description project
high school
january b
window xp
m office
india pvt
quality check
limited description
html cs
description responsibility
pune university
sql server
team member
company detail
system window
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shell script
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web application
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january bachelor
university january
information technology
description company

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Deploy

Resume Screening System

Classify resumes into job categories using Machine Learning · Built with NLP + TF-IDF

Train ML Models

🚀 Train Both Models (MNB + KNN)

✅ Models trained and saved!

94.2%
Multinomial Naive Bayes Accuracy

98.3%
KNN (OneVsRest) Accuracy

Classification Report — MNB

Classification Report — KNN

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94.2%
Multinomial Naive Bayes Accuracy

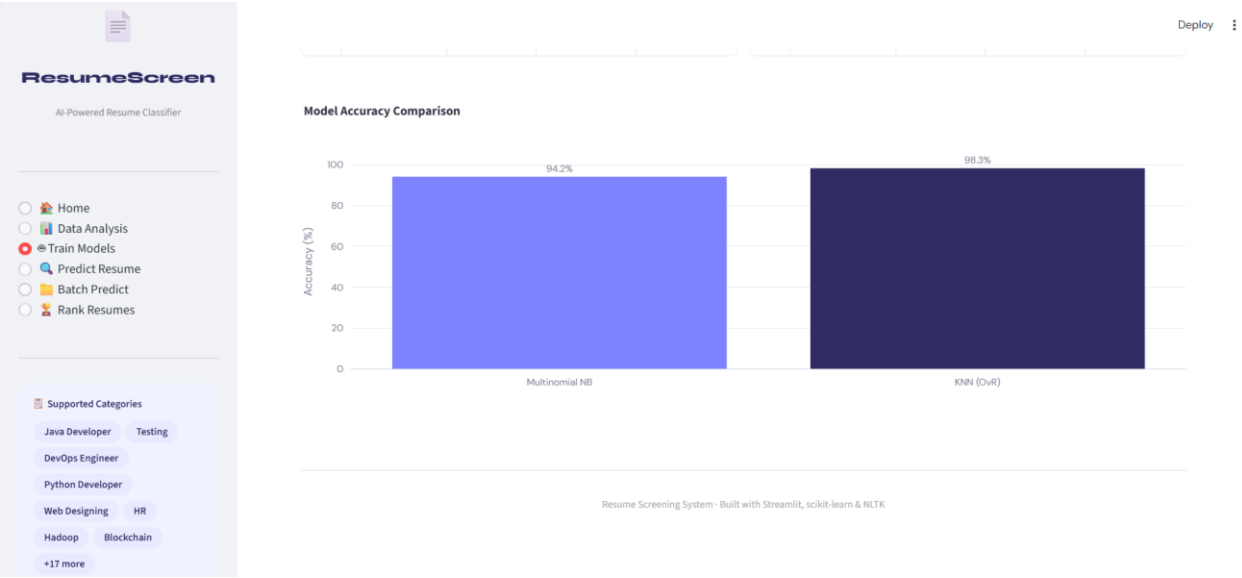
98.3%
KNN (OneVsRest) Accuracy

Classification Report — MNB

	precision	recall	f1-score	support
0	1.00	0.50	0.67	6.00
1	1.00	1.00	1.00	8.00
2	1.00	0.83	0.91	6.00
3	1.00	1.00	1.00	10.00
4	1.00	1.00	1.00	5.00
5	1.00	1.00	1.00	9.00
6	1.00	1.00	1.00	9.00
7	1.00	1.00	1.00	9.00
8	1.00	0.88	0.94	17.00
9	1.00	0.30	0.46	10.00

Classification Report — KNN

	precision	recall	f1-score	support
0	1.00	1.00	1.00	6.00
1	1.00	1.00	1.00	8.00
2	1.00	1.00	1.00	6.00
3	1.00	1.00	1.00	10.00
4	1.00	1.00	1.00	5.00
5	1.00	1.00	1.00	9.00
6	1.00	0.78	0.88	9.00
7	1.00	1.00	1.00	9.00
8	1.00	0.88	0.94	17.00
9	1.00	1.00	1.00	10.00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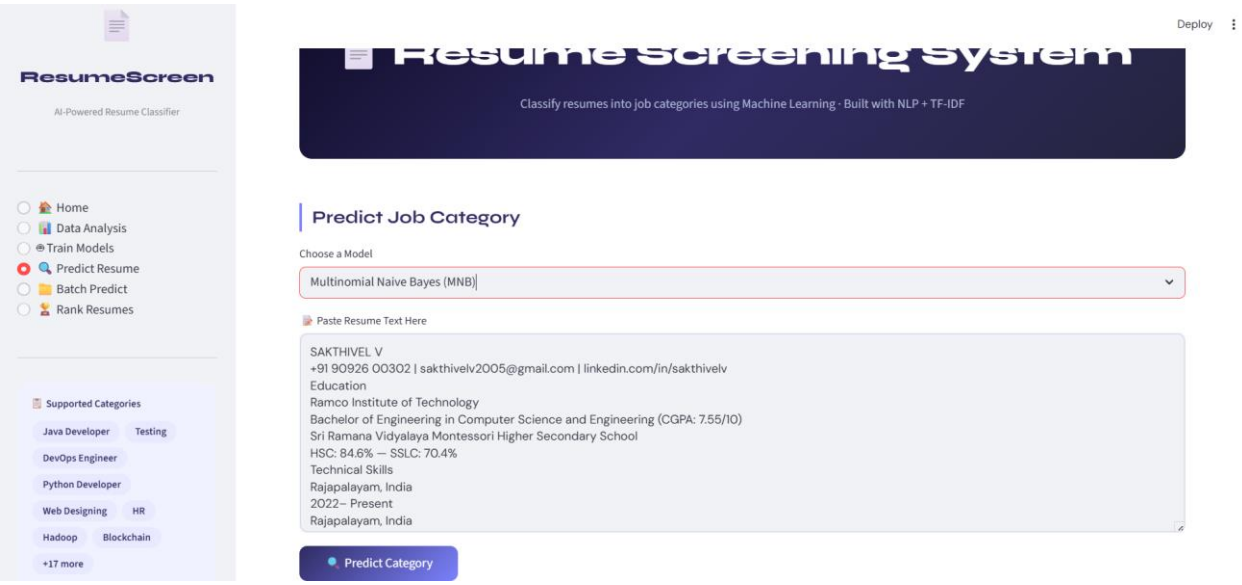
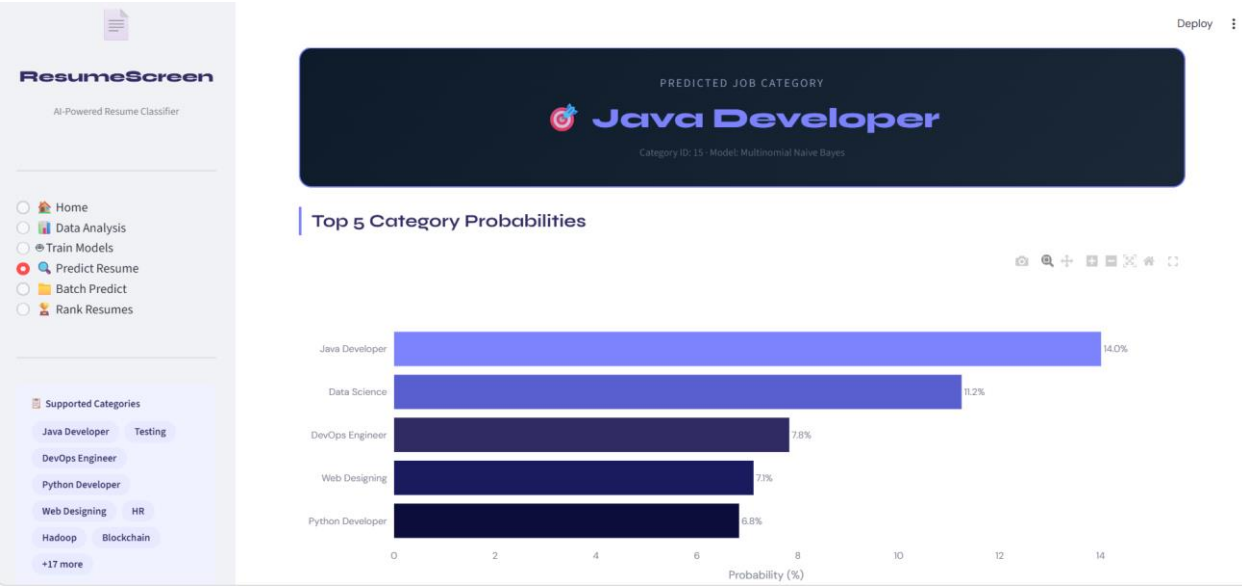
Model Accuracy Comparison
Accuracy (%)
100
80
60
40
20
0
94.2%
98.3%
Multinomial NB
KNN (OvR)
Resume Screening System · Built with Streamlit, scikit-learn & NLTK


ResumeScreen
AI-Powered Resume Classifier
Home
Data Analysis
Train Models
Predict Resume
Batch Predict
Rank Resumes
Supported Categories
Java Developer
Testing
DevOps Engineer
Python Developer
Web Designing
HR
Hadoop
Blockchain
+17 more
Deploy
Resume Screening System
Classify resumes into job categories using Machine Learning · Built with NLP + TF-IDF
Predict Job Category
Choose a Model
Multinomial Naive Bayes (MNB)
Paste Resume Text Here
SAKTHIVEL V
+91 90926 00302 | sakthivelv2005@gmail.com | linkedin.com/in/sakthivelv
Education
Ramco Institute of Technology
Bachelor of Engineering in Computer Science and Engineering (CGPA: 7.55/10)
Sri Ramana Vidyalaya Montessori Higher Secondary School
HSC: 84.6% — SSLC: 70.4%
Technical Skills
Rajapalayam, India
2022– Present
Rajapalayam, India
Predict Category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Hadoop
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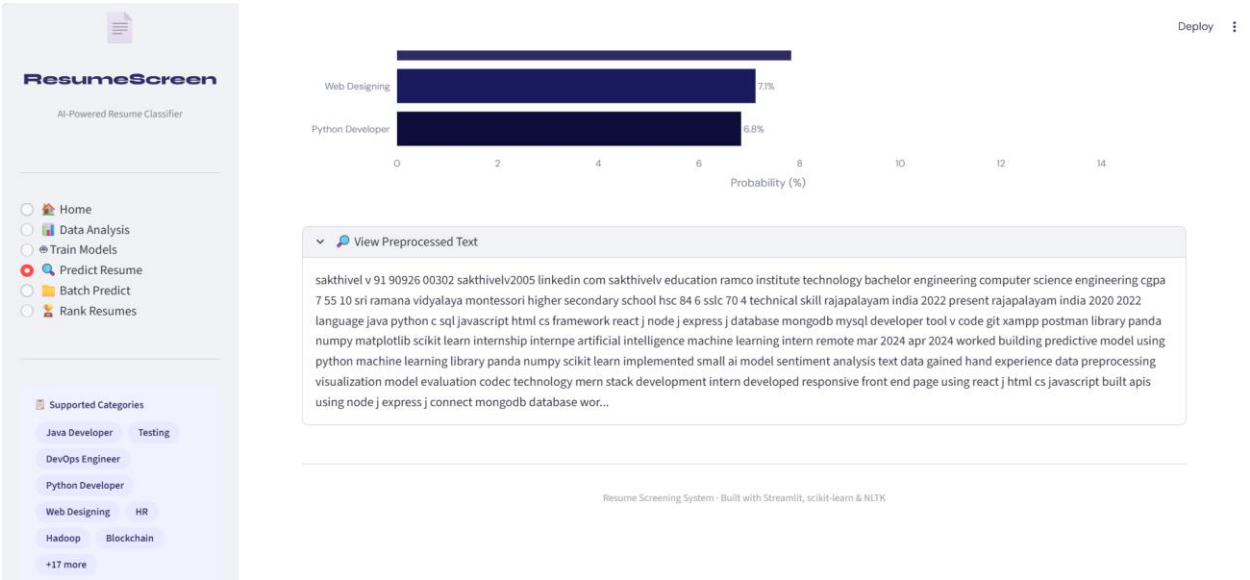
+17 more
Deploy
PREDICTED JOB CATEGORY
Java Developer
Category ID: 15 · Model: Multinomial Naive Bayes
Top 5 Category Probabilities
Java Developer
14.0%
Data Science
11.2%
DevOps Engineer
7.8%
Web Designing
7.1%
Python Developer
6.8%
0
2
4
6
8
10
12
14
Probability (%)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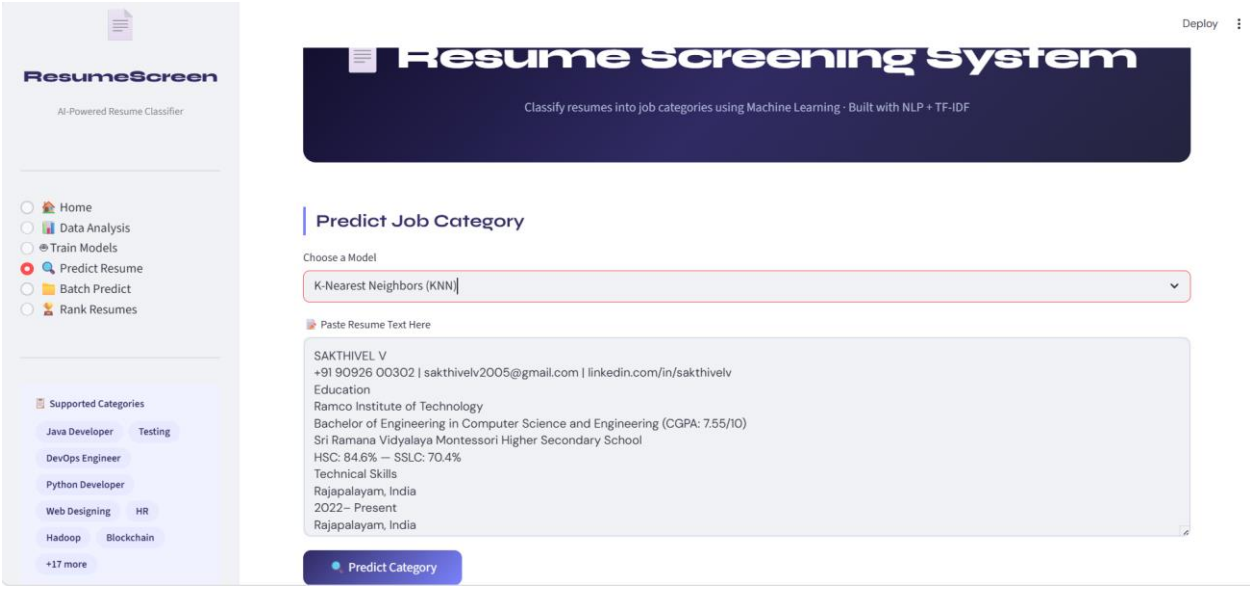
Blockchain
+17 more
Deploy
Web Designing
7.1%
Python Developer
6.8%
0
2
4
6
8
10
12
14
Probability (%)
View Preprocessed Text
sakthivel v 91 90926 00302 sakthivelv2005 linkedin com sakthivelv education ramco institute technology bachelor engineering computer science engineering cgpa 7 55 10 sri ramana vidyalaya montessori higher secondary school hsc 84 6 sslc 70 4 technical skill rajapalayam india 2022 present rajapalayam india 2020 2022 language java python c sql javascript html cs framework react j node j express j database mongodb mysql developer tool v code git xampp postman library panda numpy matplotlib scikit learn internship internpe artificial intelligence machine learning intern remote mar 2024 apr 2024 worked building predictive model using python machine learning library panda numpy scikit learn implemented small ai model sentiment analysis text data gained hand experience data preprocessing visualization model evaluation codec technology mern stack development intern developed responsive front end page using react j html cs javascript built apis using node j express j connect mongodb database wor...
Resume Screening System · Built with Streamlit, scikit-learn & NLTK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ResumeScreen

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- Batch Predict
- Rank Resumes

Supported Categories

Java Developer | Testing | DevOps Engineer | Python Developer | Web Designing | HR | Hadoop | Blockchain | +17 more

PREDICTED JOB CATEGORY

Data Science

Category ID: 6 · Model: KNN (OneVsRest)

View Preprocessed Text

sakthivel v 91 90926 00302 sakthivelv2005 linkedin com sakthivelv education ramco institute technology bachelor engineering computer science engineering cgpa 7 55 10 sri ramana vidyalaya montessori higher secondary school hsc 84 6 sslc 70 4 technical skill rajapalayam india 2022 present rajapalayam india 2020 2022 language java python c sql javascript html cs framework react j node j express j database mongodb mysql developer tool v code git xampp postman library panda numpy matplotlib scikit learn internship internpe artificial intelligence machine learning intern remote mar 2024 apr 2024 worked building predictive model using python machine learning library panda numpy scikit learn implemented small ai model sentiment analysis text data gained hand experience data preprocessing visualization model evaluation codec technology mern stack development intern developed responsive front end page using react j html cs javascript built apis using node j express j connect mongodb database wor...

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Batch Resume Prediction

Upload a CSV file with a `Resume` column to classify multiple resumes at once.

Choose Model for Batch Prediction

Multinomial Naive Bayes (MNB)

Upload CSV

ResumeDataSet.csv 3.0MB

Loaded 962 resumes.

	Category	Resume
0	Data Science	Skills * Programming Languages: Python (pandas, numpy, scipy, scikit-learn, matplotlib), Sql, Java, JavaScript/JQuery. * Mach
1	Data Science	Education Details May 2013 to May 2017 B.E UIT-RGPV Data Scientist Data Scientist - Matelabs Skill Details Python- Exprienc
2	Data Science	Areas of Interest Deep Learning, Control System Design, Programming in-Python, Electric Machinery, Web Development, Analy
3	Data Science	Skills âØ¢ R âØ¢ Python âØ¢ SAP HANA âØ¢ Tableau âØ¢ SAP HANA SQL âØ¢ SAP HANA PAL âØ¢ MS SQL âØ¢ SAP Lumira âØ¢ (
4	Data Science	Education Details MCA YMCAUST, Faridabad, Haryana Data Science internship Skill Details Data Structure- Exprience - Les

Run Batch Prediction

Deploy

ResumeScreen

AI-Powered Resume Classifier

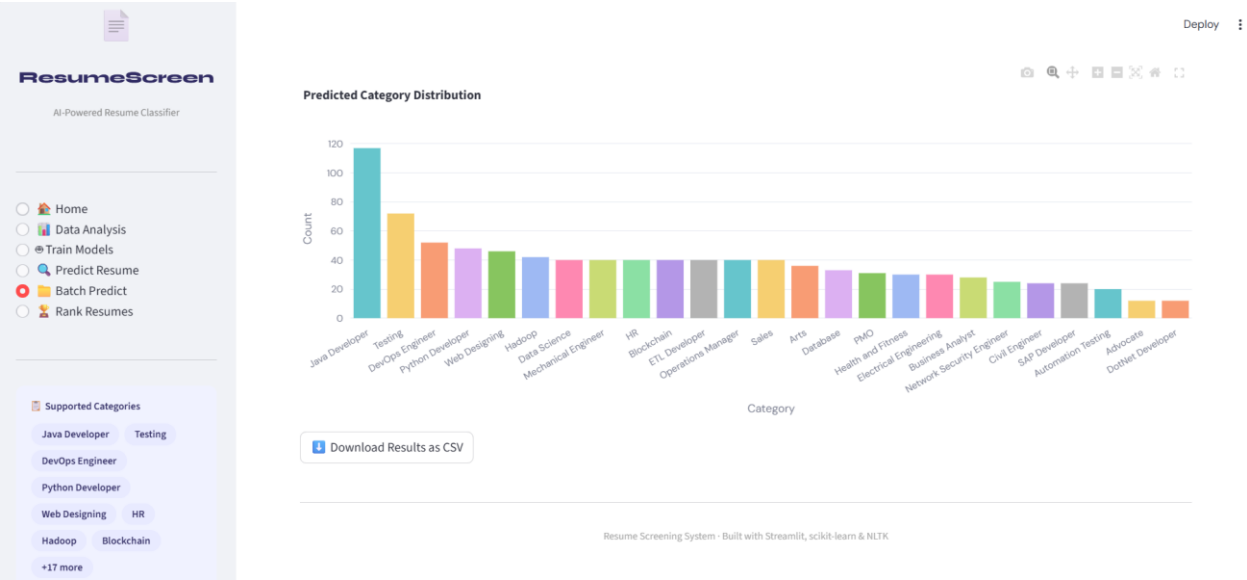
- Home
- Data Analysis
- Train Models
- Predict Resume
- Batch Predict
- Rank Resumes

Supported Categories

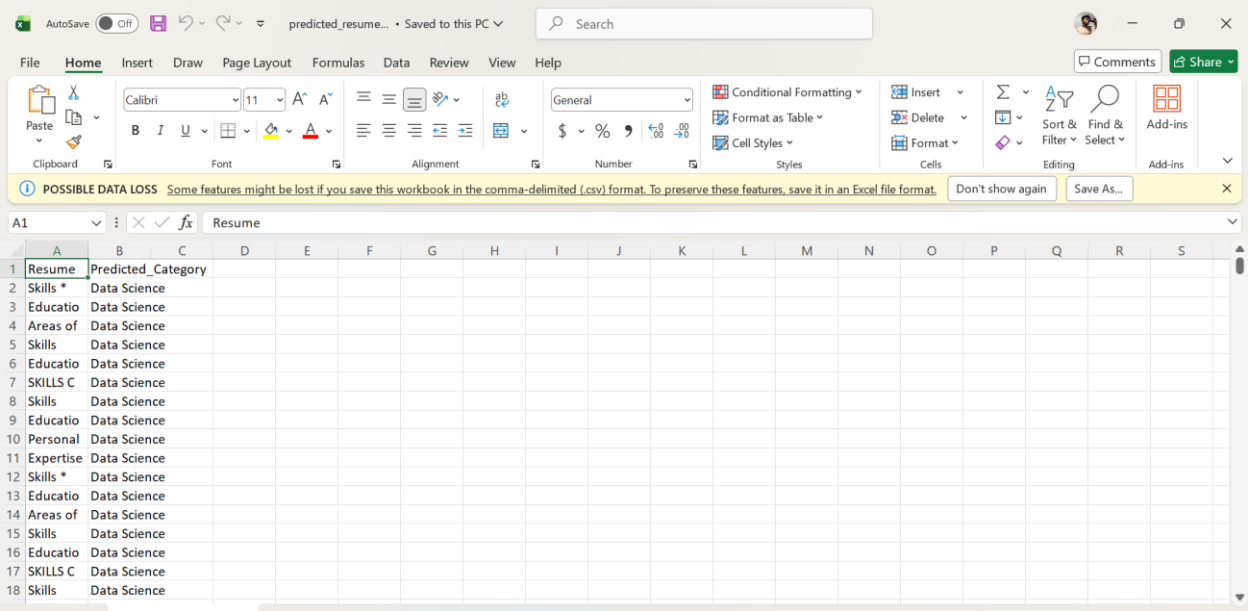
Java Developer | Testing | DevOps Engineer

Batch prediction complete!

	Resume	Predicted_Category
0	Skills * Programming Languages: Python (pandas, numpy, scipy, scikit-learn, matplotlib), Sql, Java, JavaScript/JQuery. * M	Data Science
1	Education Details May 2013 to May 2017 B.E UIT-RGPV Data Scientist Data Scientist - Matelabs Skill Details Python- Expri	Data Science
2	Areas of Interest Deep Learning, Control System Design, Programming in-Python, Electric Machinery, Web Development, A	Data Science
3	Skills âØ¢ R âØ¢ Python âØ¢ SAP HANA âØ¢ Tableau âØ¢ SAP HANA SQL âØ¢ SAP HANA PAL âØ¢ MS SQL âØ¢ SAP Lumira â	Data Science
4	Education Details MCA YMCAUST, Faridabad, Haryana Data Science internship Skill Details Data Structure- Exprience -	Data Science
5	SKILLS C Basics, IOT, Python, MATLAB, Data Science, Machine Learning, HTML, Microsoft Word, Microsoft Excel, Microsoft P	Data Science
6	Skills âØ¢ Python âØ¢ Tableau âØ¢ Data Visualization âØ¢ R Studio âØ¢ Machine Learning âØ¢ Statistics IABAC Certified Da	Data Science
7	Education Details B.Tech Rayat and Bahra Institute of Engineering and Biotechnology Data Science Data Science Skill De	Data Science
8	Personal Skills âØ¢ Ability to quickly grasp technical aspects and willingness to learn âØ¢ High energy levels & Result orien	Data Science
9	Expertise âØØ Data and Quantitative Analysis âØØ Decision Analytics âØØ Predictive Modeling âØØ Data-Driven Personaliz	Data Science



Downloaded Results as CSV :



Resume	Predicted_Category
Skills *	Data Science
Educatio	Data Science
Areas of	Data Science
Skills	Data Science
Educatio	Data Science
SKILLS C	Data Science
Skills	Data Science
Educatio	Data Science
Personal	Data Science
Expertise	Data Science
Skills *	Data Science
Educatio	Data Science
Areas of	Data Science
Skills	Data Science
Educatio	Data Science
SKILLS C	Data Science
Skills	Data Science

ResumeScreen

AI-Powered Resume Classifier

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Supported Categories

Java Developer | Testing | DevOps Engineer | Python Developer | Web Designing | HR | Hadoop | Blockchain | +17 more

Deploy

Resume Screening System

Classify resumes into job categories using Machine Learning · Built with NLP + TF-IDF

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Upload a CSV file with a `Resume` column to classify multiple resumes at once.

Choose Model for Batch Prediction

K-Nearest Neighbors (KNN)

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ResumeDataSet.csv 3.0MB

Loaded 962 resumes.

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Run Batch Prediction

Batch prediction complete!

ResumeScreen

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Batch prediction complete!

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5	SKILLS C Basics, IOT, Python, MATLAB, Data Science, Machine Learning, HTML, Microsoft Word, Microsoft Excel, Microsoft P	Data Science
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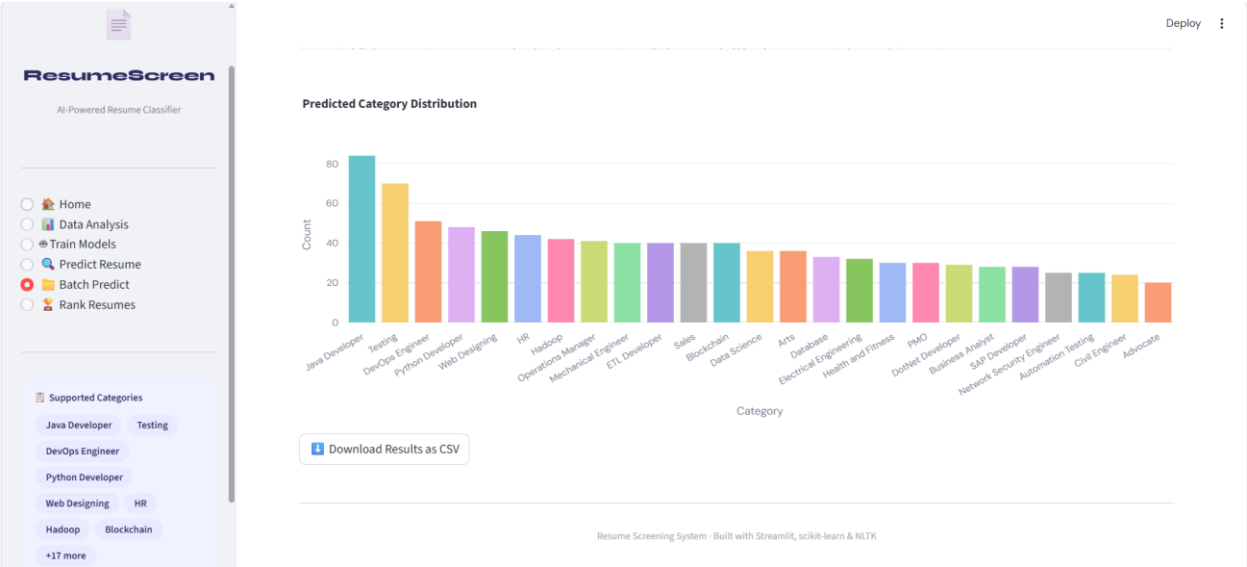
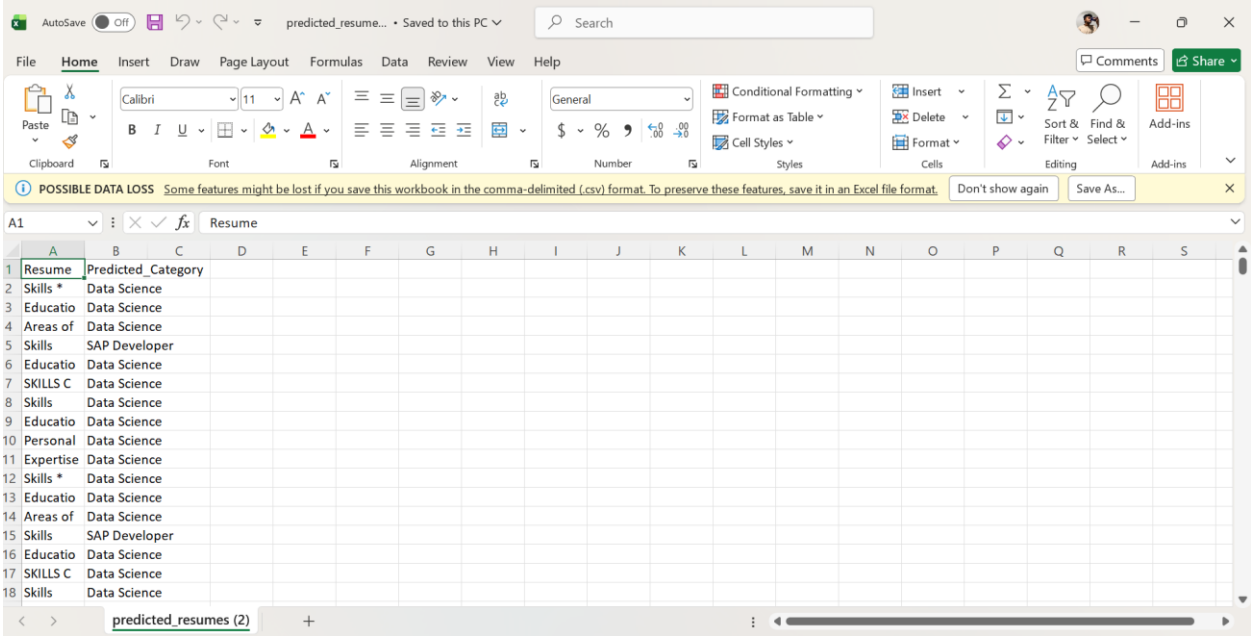
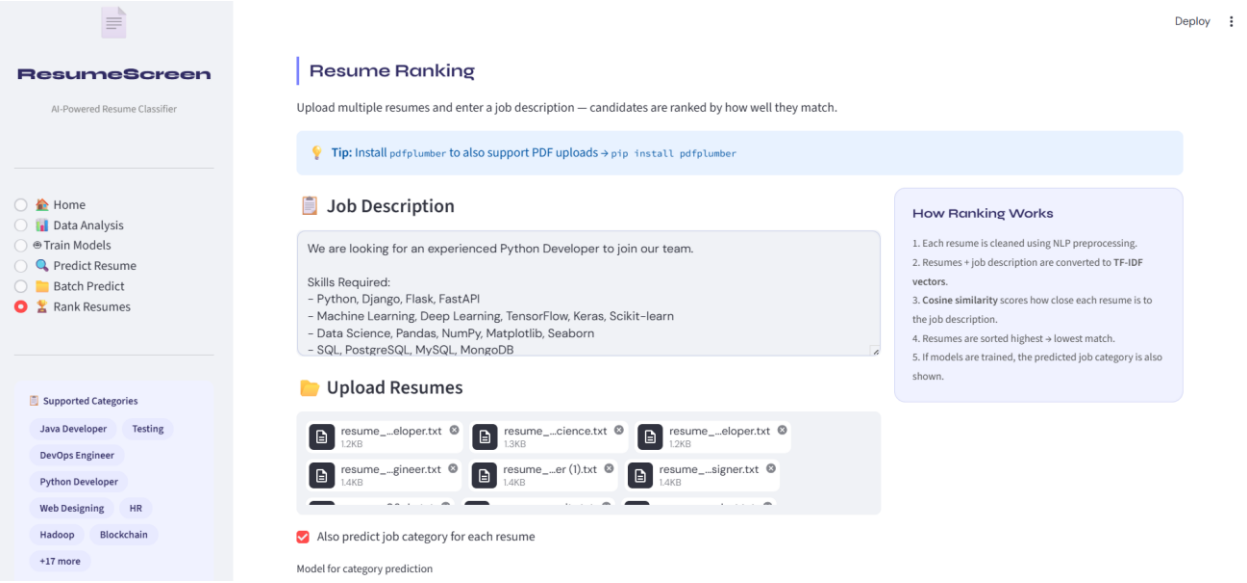
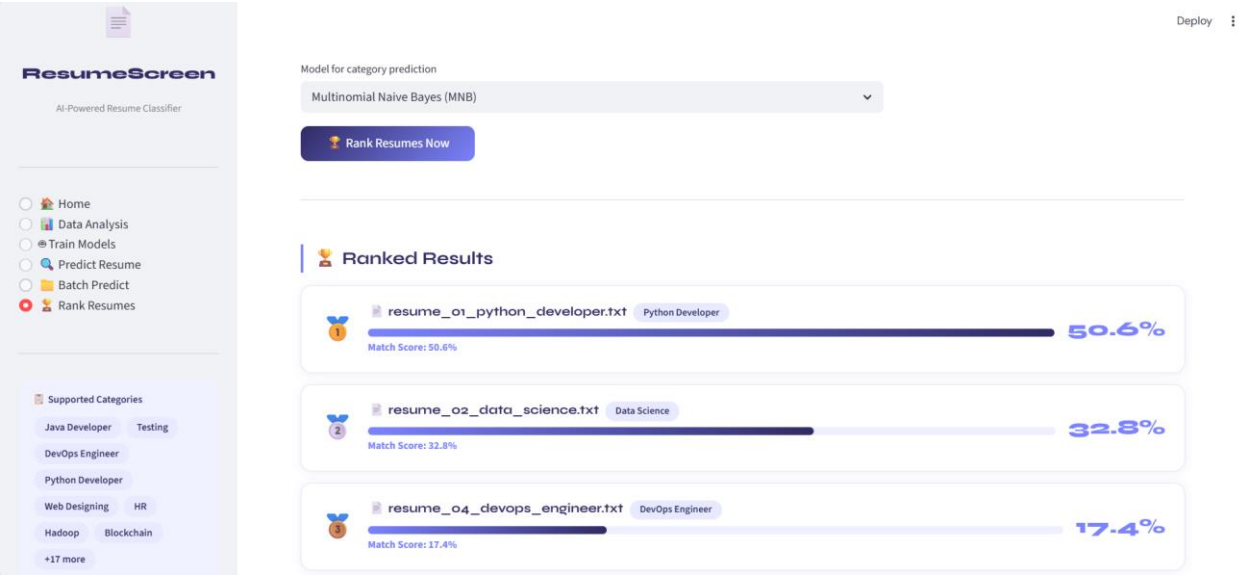
Predicted Category Distribution





Resume	Predicted_Category
Skills *	Data Science
Educatio	Data Science
Areas of	Data Science
Skills	SAP Developer
Educatio	Data Science
SKILLS C	Data Science
Skills	Data Science
Educatio	Data Science
Personal	Data Science
Expertise	Data Science
Skills *	Data Science
Educatio	Data Science
Areas of	Data Science
Skills	SAP Developer
Educatio	Data Science
SKILLS C	Data Science
Skills	Data Science

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HR
Hadoop
Blockchain
+17 more
Deploy
Model for category prediction
Multinomial Naive Bayes (MNB)
Rank Resumes Now
Ranked Results
resume_01_python_developer.txt
Python Developer
Match Score: 50.6%
50.6%
resume_02_data_science.txt
Data Science
Match Score: 32.8%
32.8%
resume_04_devops_engineer.txt
DevOps Engineer
Match Score: 17.4%
17.4%

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+17 more
Deploy
#4
resume_03_java_developer.txt
Java Developer
Match Score: 9.5%
9.5%
#5
resume_09_dotnet_developer.txt
Java Developer
Match Score: 7.1%
7.1%
#6
resume_10_testing_qa.txt
Testing
Match Score: 6.3%
6.3%
#7
resume_08_business_analyst.txt
Business Analyst
Match Score: 3.7%
3.7%
#8
resume_05_web_designer (1).txt
Web Designing
Match Score: 3.0%
3.0%

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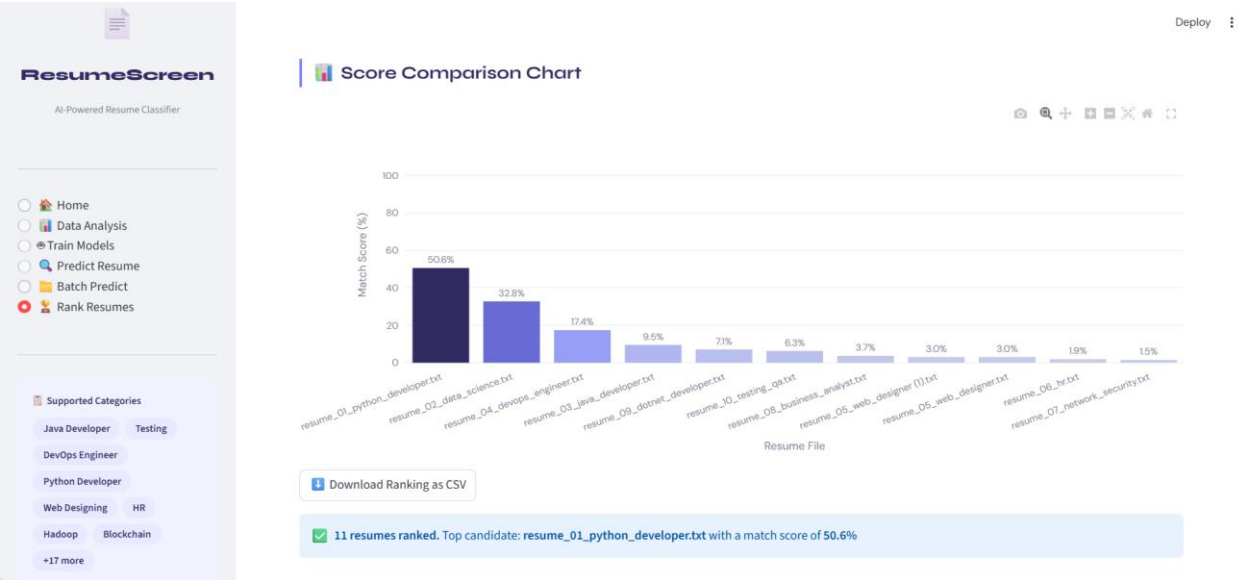
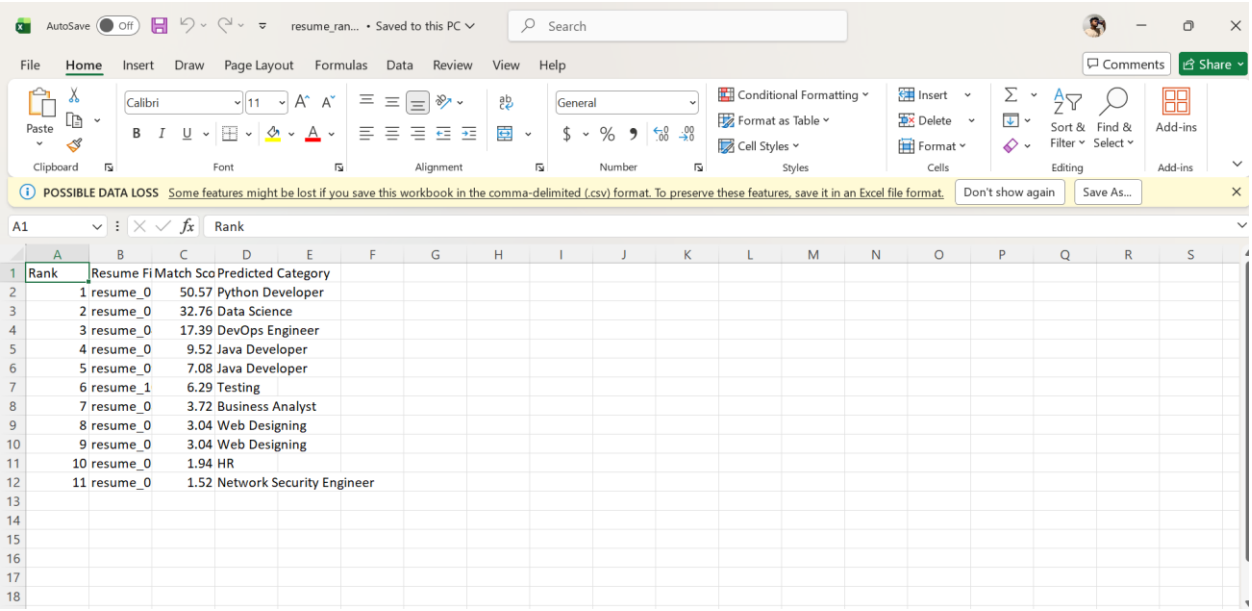
#9
resume_05_web_designer.txt
Web Designing
Match Score: 3.0%
3.0%
#10
resume_06_hr.txt
HR
Match Score: 1.9%
1.9%
#11
resume_07_network_security.txt
Network Security Engineer
Match Score: 1.5%
1.5%



Downloaded Ranked CSV



Rank	Resume Fi	Match Sco	Predicted Category
1	resume_0	50.57	Python Developer
2	resume_0	32.76	Data Science
3	resume_0	17.39	DevOps Engineer
4	resume_0	9.52	Java Developer
5	resume_0	7.08	Java Developer
6	resume_1	6.29	Testing
7	resume_0	3.72	Business Analyst
8	resume_0	3.04	Web Designing
9	resume_0	3.04	Web Designing
10	resume_0	1.94	HR
11	resume_0	1.52	Network Security Engineer

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Supported Categories

Java Developer | Testing | DevOps Engineer | Python Developer | Web Designing | HR | Hadoop | Blockchain | +17 more

Deploy

Resume Ranking

Upload multiple resumes and enter a job description — candidates are ranked by how well they match.

Tip: Install `pdfplumber` to also support PDF uploads → `pip install pdfplumber`

Job Description

We are looking for an experienced Python Developer to join our team.

Skills Required:
- Python, Django, Flask, FastAPI
- Machine Learning, Deep Learning, TensorFlow, Keras, Scikit-learn
- Data Science, Pandas, NumPy, Matplotlib, Seaborn
- SQL, PostgreSQL, MySQL, MongoDB

How Ranking Works

1. Each resume is cleaned using NLP preprocessing.
2. Resumes + job description are converted to **TF-IDF vectors**.
3. **Cosine similarity** scores how close each resume is to the job description.
4. Resumes are sorted highest → lowest match.
5. If models are trained, the predicted job category is also shown.

Upload Resumes

resume_...eloper.txt 1.2KB | resume_...cience.txt 1.3KB | resume_...eloper.txt 1.2KB | resume_...gineer.txt 1.4KB | resume_...er (1).txt 1.4KB | resume_...signer.txt 1.4KB

Also predict job category for each resume

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Deploy

Model for category prediction

K-Nearest Neighbors (KNN)

Rank Resumes Now

Ranked Results

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#4	resume_03_java_developer.txt	Java Developer	9.5%
#5	resume_09_dotnet_developer.txt	DotNet Developer	7.1%
#6	resume_10_testing_qa.txt	Automation Testing	6.3%
#7	resume_08_business_analyst.txt	Business Analyst	3.7%
#8	resume_05_web_designer (1).txt	Web Designing	3.0%

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Supported Categories

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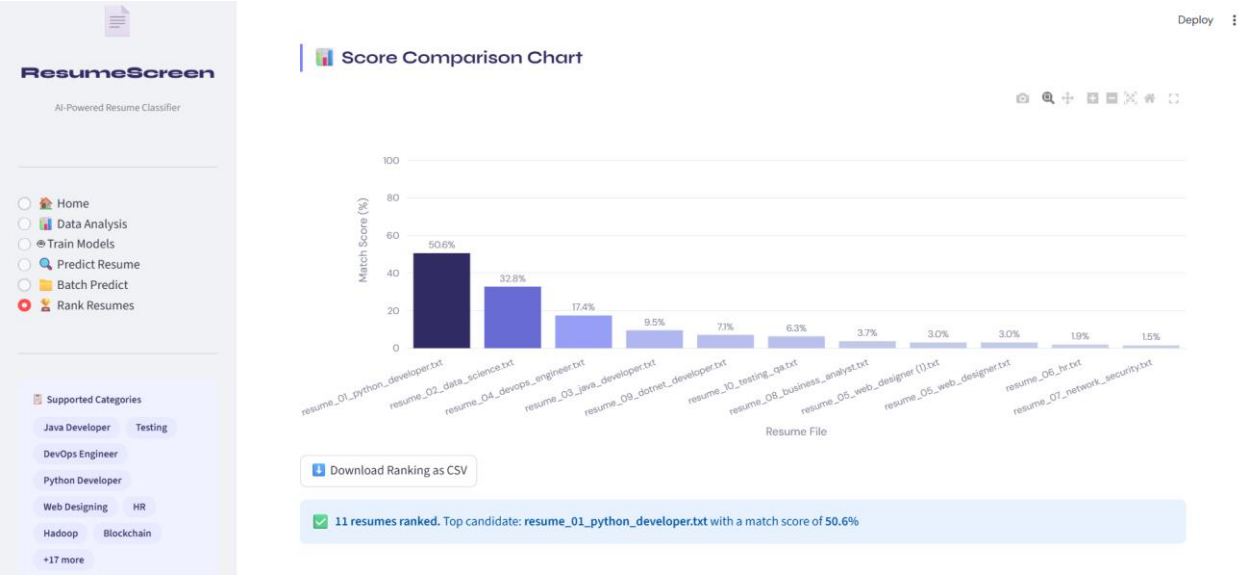

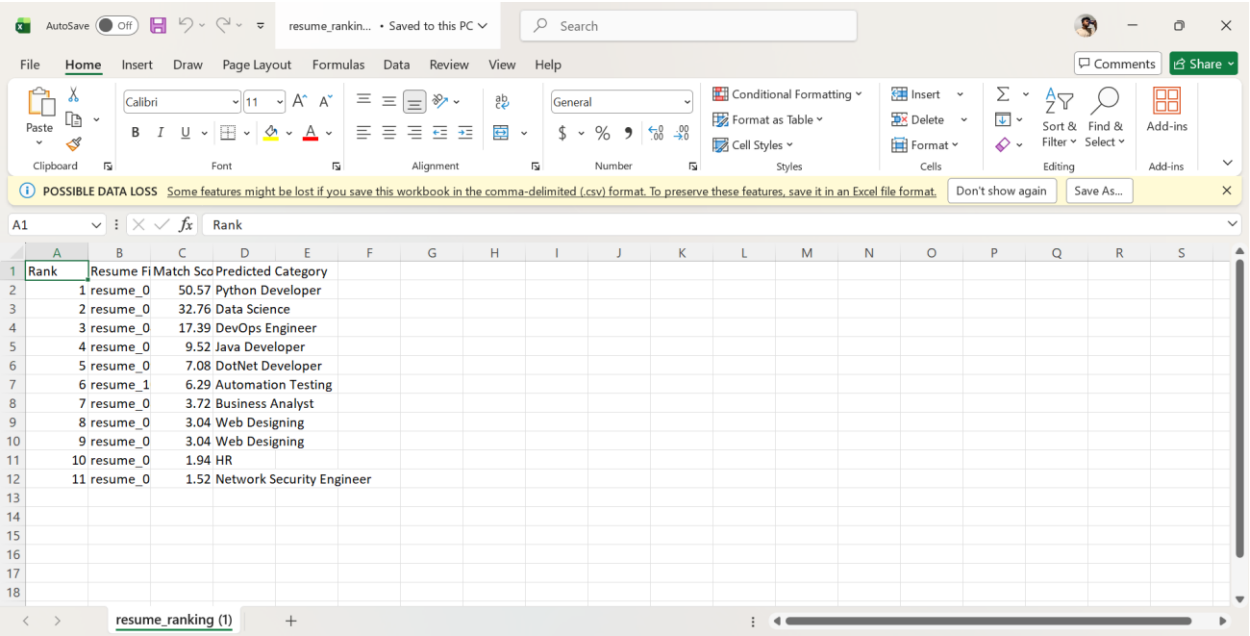

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6	resume_1	6.29	Automation Testing
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10	resume_0	1.94	HR
11	resume_0	1.52	Network Security Engineer

CHAPTER 6

CONCLUSION & FUTURE ENHANCEMENT

6.1 Conclusion

The Resume Screening System using NLP successfully automates the recruitment process by extracting relevant skills and ranking candidates efficiently. It reduces manual effort and improves accuracy in candidate selection. This project demonstrates how NLP techniques can be applied in real-world HR applications to improve productivity and decision-making.

6.2 Future Enhancements

The current Resume Screening System provides an efficient solution for ranking candidates using NLP techniques. However, several enhancements can be implemented to further improve its performance, accuracy, and usability.

- **Use of Advanced NLP Models:**
  The system currently relies on TF-IDF, which is a bag-of-words approach and does not capture the context of words. In the future, advanced NLP models such as BERT or other transformer-based models can be integrated to understand semantic meaning and context more effectively, leading to more accurate matching between resumes and job descriptions.
- **Enhanced User Interface:**
  Although the system uses Streamlit for the interface, it can be further improved by designing a more interactive and visually appealing GUI. Features such as dashboards, filters, search options, and better visualization of candidate scores can enhance user experience and usability.
- **Extraction of Detailed Information:**
  The system can be extended to extract more structured information from resumes, such as educational qualifications, years of experience, certifications, and previous job roles. This would allow more refined filtering and better decision-making in the recruitment process.
- **Multi-language Support:**
  Currently, the system primarily supports English resumes. Future enhancements can include multi-language processing capabilities, enabling the system to analyze resumes written in different languages using language detection and translation techniques.
- **Integration with Job Portals:**
  The system can be integrated with online job portals or recruitment platforms to automatically fetch resumes and job descriptions, making the screening process fully automated.
- **Real-time Processing and Scalability:**
  The system can be optimized to handle large-scale data in real time by deploying it on cloud platforms. This would make it suitable for enterprise-level applications with high volumes of resumes.